



Polypropylene nanocomposite fibers: A review of current trends and new developments

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Abstract

In this review, traditional and novel techniques for producing micro- and nano- fibers are discussed and various nanofillers, their modifications and polypropylene (PP) functionalization routes are presented. Their influence on PP properties is discussed and new PP composite fiber applications are presented. This review reveals interesting conclusions, such as that in terms of mechanical reinforcement, there is no nano-filler that can improve tensile strength to the extent that it is improved by drawing. However, in some cases, composite drawn fibers are characterized by higher tensile strength than drawn neat PP. With some notable exceptions, the PP nanocomposites lack of “dramatic” properties improvement is mainly due to the non-polar nature of the hydrocarbon chain, which does not favor strong intermolecular interactions with most popular (mainly inorganic) nano-fillers. However, other properties such as electric conductivity, water contact angle and others can be effectively altered using various nanofillers in PP matrices.

Keywords

Polypropylene, nanocomposite, fiber, drawing, additive

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Introduction

Isotactic Polypropylene (i-PP) was first developed in 1953 by Professor Giulio Natta, who continued Karl Ziegler's research on organometallic catalysts for stereospecific polymerization, thus discovering new polymers with a sterically ordered structure. Professors Natta and Ziegler were honored with The Nobel Prize in Chemistry, in 1963, "for their discoveries in the field of the chemistry and technology of high polymers". The industrial production of isotactic polypropylene began in 1957.¹

The global plastic production has increased almost sevenfold, from 50 Mt in 1976 to 322 Mt in 2015. In 2018, the global plastics production was almost 360 Mt. The same year, the production in Europe was about 62 Mt, providing jobs to more than 1.6 million employees at almost 60.000 companies and a turnover more than 360 billion euros. Polypropylene (PP) is the second most produced plastic after polyethylene (PE). In 2015, about 62 Mt of PP resin and 30 Mt of PP fibers were produced worldwide, while in 2018, the PP resin demand in Europe was almost 10 Mt.^{2–6} Combined with the projection that the global PP production is expected to reach around 88 Mt by 2026,⁷ the above shows how important PP is in the global economy.

PP is a commonly used thermoplastic. It presents high melting temperature (160–166°C for isotactic PP), low density (0.898–0.908 g/cm³ for isotactic PP), high chemical inertness, the capability to be produced in different structures and properties, via various processes, and it can be modified by adding reinforcing agents, fillers or other polymers and, also, it can be grafted with various functional groups.⁸ Thus, PP has a higher diversity in its applications, among all thermoplastics.

Packaging is a major application for polypropylene. PP films less than 10 mils (254 µm) thick can be produced via an extrusion process. Due to their flexibility, resistance to moisture and chemical stability, PP films are used as flexible packaging materials for food, clothing and tobacco. PP sheets about 40 mils (~ 1 mm) thick, are also produced by an extrusion process. Followed by a thermoforming process, PP sheets can be formed into a desired shape and can be used as rigid packaging containers, suitable to transport and store various products. Rigid packaging containers can also be produced via an injection molding process. Bottles as well as toys, sports equipment, garden furniture, screw caps and disposable syringes are some products that are manufactured via the injection molding process. The blow molding process is appropriate, mainly, for producing water bottles, various containers and jars. Taking into account its low price, PP tends to replace many packaging materials, such as paper, wood and metal. PP is also used widely in the automotive industry, due to its light weight and long-life. The high thermal stability of PP makes it suitable to produce food containers used in microwave ovens, or for medical parts that have to be frequently sterilized, which require relatively high temperature treatments.⁹

However, in 2015, almost 33% of the global PP production was for fibers.² Many products for home, agriculture, clothing and packaging applications are produced using PP fibers.⁹ PP fibers are also used in the construction sector, mainly, to improve concrete strength (fiber-reinforced concrete).¹⁰ During the last decade, much research has been conducted on nano-scaled PP fibers. It was found that they are very promising materials for many applications, such as water filtration, or as substrates for sensor development. PP fibers can be produced via several methods such as melt blowing, melt spinning with subsequent fiber stretching and solution or melt electrospinning. Electrospinning, is the most widely used technique to produce PP nanofibers.

Polypropylene structure and properties

In general, compared to other common plastics, such as PE, PP is characterized by low density and high thermal stability, as it can tolerate higher continuous use temperatures (e.g. 100 °C versus 55 °C for high density PE).¹¹ It offers good processability via extrusion and injection molding. However other polymer processing techniques, such as thermoforming, vacuum forming, and blow molding are more challenging as they require newly developed resins. In addition to that, other PP disadvantages are its relatively high thermal expansion (but still lower compared to low and high density PE) and high mold shrinkage. PP exhibits low impact strength at low (near or below zero °C) temperatures, poor ultraviolet (UV) resistance and poor oxidative resistance in presence of copper and other metals.^{11,12}

Such properties, as well as the most important thermal and mechanical properties of polypropylene arise from its structure. Homopolymer polypropylene presents three different tacticities:

- isotactic-PP (i-PP)
- syndiotactic-PP (s-PP)
- atactic-PP (a-PP)

As a polyolefin, it presents a hydrocarbon chain, without polar groups. Therefore, polypropylene chains interact solely by dispersion forces, and present very low reactivity. Consequently, it is chemically resistant and cannot be dissolved in common organic solvents. Nevertheless, xylene and petrol ether (mixture of pentane and hexane) have been used to extract some soluble PP fractions. However, thermal, and mechanical properties, as well as its behavior during polymer processing, strongly depend on its ability to form crystals, which is directly connected to its stereochemical structure. Thus, the type/tacticity of PP is considered of major importance.¹³ In general, i-PP is semicrystalline with degrees of crystallinity between 40–70%, which depend on the degree of tacticity.¹² Both i-PP and s-PP give sharp peaks in X-Ray characterization methods (e.g. X-Ray diffraction), while atactic PP presents a very broad peak in XRD, characteristic of amorphous materials.

Crystallinity in neat PP

Isotactic PP, which is mainly used to produce fibers, exhibits polymorphism (α -, β - and γ - crystals and a mesomorphic/smectic form). The α -form is the dominant i-PP form. In more detail, there are two α -forms, the α_1 - and α_2 -forms, which have different space group symmetries. The α_2 -form is slightly more thermodynamically favorable and is associated with high annealing temperatures.¹⁴ The β -form crystals present lower density, higher crystallization rate,^{15,16} lower melting point and metastability compared to the α -form.¹⁷ Upon heating, β -form converts to the α -form,¹³ a phenomenon known as $\beta\alpha$ -recrystallization. In DSC measurements the β -form endothermic melting peak, the $\beta\alpha$ -exothermic recrystallization peak and the α -form endothermic melting peak partially overlap. Also, the α -form melting peak which is originally present, and the α -form melting peak produced by $\beta\alpha$ -recrystallization can be detected.¹⁸

Various additives can be used to facilitate the β -crystal formation. However, β -crystallization is possible without proper additives under shear conditions.^{17,19} Also, it is favored when i-PP crystallizes between 115 and 135 °C.²⁰ The i-PP β -form was the first polymeric crystallographic structure that was recognized to be frustrated (that is a crystalline structure where atoms/molecules are organized locally but not with strict periodicity).²¹ The higher β -form crystallization rate, compared to the α -form, occurs below 140 °C,^{15,16} while at higher isothermal crystallization temperatures, the α -form crystals grow faster.²² The β -nucleated PP properties and processability have been recently reviewed.²³ The γ -form is uncommon under typical processing conditions. It is characterized by similar, or slightly higher density, than the α -form. A transition to α -form may occur, but it depends on the supercooling history (during γ -form crystallization). γ -form is associated with low molecular weight PP crystallization. Also, γ -form crystallization is favored when there are many defects (i.e. low i-PP content). High pressure crystallization favors the γ -form, even in high molecular weight, or in PP with high tacticity.²⁴

The mesomorphic PP (or smectic PP) is favored under very rapid cooling (is the predominant form for cooling rates >80 °C/sec).²⁵ For this reason, PP filaments under tensile stress mostly form α -crystals, when quenched in air, and mesomorphic structure, if rapidly quenched in an ice-water bath.²⁵ Mesomorphic structure is an intermediate state of order between crystalline and amorphous states and it exhibits a low temperature (65–120 °C) broad exothermic, due to smectic to α -form transition.²⁴ It has a particular commercial interest, since mesomorphic PP quenched films are characterized by low density and transparency. However, smectic PP is also important in fiber manufacturing, since it is the initially desired form after spinning and prior to drawing.

The two most common i-PP crystal forms (α - and β -) have differences in mechanical properties, besides differences in melting temperature and crystallization rate. The β -form has higher impact strength.²⁰ Such improved mechanical properties are attributed to the $\beta\alpha$ -recrystallization and the associated energy

dissipation during yield. More specifically, the β -form is metastable, as it is kinetically stable and not thermodynamically stable. Although recrystallization is an exothermic phenomenon, polymer chains need energy in order to rearrange. Instead of heat, tensile strength can provide the required energy and, thus, during yield energy dissipation occurs. Also, it has been suggested that the main factor for improved β -phase mechanical properties is the amorphous phase connection between the β -phase crystallites.²⁶ However, it is difficult to obtain β -form in fibers due to the $\beta\alpha$ -transition during stretching.²⁰

s-PP also exhibits polymorphism. The metastable form II was first reported in the low stereoregular s-PP oriented fibers.²⁷ The s-PP stable form I, which is favored under common processing/crystallization conditions, was reported two decades after the form II discovery.²⁸ The form III is obtained from highly stereoregular s-PP at high deformations by room temperature stretching.²⁹ The same research group reported the form IV, which can be obtained by exposing form III s-PP to solvent vapors.³⁰ By quenching melted s-PP to 0 °C, mesomorphic s-PP is obtained.^{31,32} Transformations from one form to another are possible. The s-PP structure, physical properties and transformations were reviewed.³³

Finally, propylene random copolymers with ethylene are characterized by lower crystallinity and lower melting point and are typically used in thermoforming and injection molding applications.¹¹

Crystallinity in PP composites and fibers

In this section, the nucleation effect of various additives on i-PP crystallinity and how this effect can also influence the fiber formation, in i-PP composite fibers, is briefly discussed and recent progress is reviewed.

Not all inorganic fillers used in PP composites and act as nucleating agents promote the same crystal structure. For example, talc and CaCO_3 are two common inorganic fillers for i-PP. Talc promotes α - crystals, while CaCO_3 promotes β - crystals.³⁴ Literature studies rarely shed light on the nucleation mechanism induced by additives and fillers and most often they only present observations about the produced crystal structure. There is special research on developing β -phase nucleating agents for i-PP. In this direction, additives, such as a complex of calcium and lanthanum with organic ligands³⁵ and N,N'-dicyclohexyl-2,6-naphthalenedicarboxamide,³⁶ have been suggested and recently reviewed.²³ Furthermore, polymer nanofibers, mainly produced by electrospinning, act as nucleating agents for i-PP. Various polymer nanofibers (e.g. polytetrafluoroethylene (PTFE), polyethylene terephthalate (PET), polyvinyl alcohol (PVA), polyamide 6/nylon 6 (PA6)) favor α -crystals,^{37,38} while isotactic polystyrene (i-PS) induces the β -crystallization in i-PP.^{39,40} Montmorillonite (MMT) modified with hexadecyl trimethyl ammonium bromide, is reported to favor α -form i-PP.²⁵

During fiber spinning and drawing, both shear and tensile stresses exist and, thus, competitive phenomena can occur regarding which i-PP crystalline structure forms. Shear stress enhances the formation of β -crystals,¹⁷ while tensile stress

favors the α -form and β -recrystallization and,²⁰ consequently, from crystallographic point of view, the α -form prevails in oriented fibers.¹³ Moreover, when the β -form i-PP is stretched at room temperature, the smectic form appears in the fibers, while for drawing temperatures above 60 °C, or fibers drawn directly from the melt, the α -form is predominant and only small β -form portions can be present.¹³ In the necking region, as elongation increases, the β -form content decreases.³⁴ If a nucleation agent is present, there is an increasing complexity regarding how the negative or positive synergy of such factors affect the final fiber crystallinity. As mentioned, some additives favor the α -form and others favor the β -form crystal structure. Moreover, β -form crystal growth rate is higher,¹⁶ than the respective α -form, for a wide crystallization temperature range. Finally, increased i-PP crystallinity prior to drawing is not desired, since it results in inhomogeneous polymer matrix.⁴¹

Therefore, using a nucleation agent most likely will require alterations in the spinning and drawing process. For example, when modified MMT was added, it promoted increased crystallinity in composite filaments and, thus, higher drawing temperature was required, compared to the neat PP.²⁵ However, for another MMT modification, it was reported that MMT acted not as a nucleation agent, but as a processing additive resulting in increased composite fiber drawability.⁴²

Polypropylene fibers

Fibers production methods

Drawing and molecular orientation. Fibers are characterized by flexibility and high length to thickness ratio. Traditionally, fibers have a 10–50 μm diameter and a 1 cm minimum length. Besides natural and synthetic polymer (organic) fibers, inorganic fibers can also be prepared e.g. glass wool. Carbon nanotubes are among the most novel fibers. In the fiber/textile science and technology the linear density is commonly used instead, or additionally to the diameter. The linear density unit of measurement is tex (1tex = 1 g/km) and dtex (g/10 km) which closely relates to the Denier unit (1 denier = 1 g/9000 m). Polymer fibers according to their strength and stability can be classified in four categories:⁴¹

- fibers (natural or regenerated) of medium strength and breaking elongation
- synthetic fibers characterized by high toughness
- high modulus and high tenacity fibers
- elastomeric fibers.

From chemical point of view there are mainly six fiber types:⁴¹

- cellulosic (e.g. cotton)
- protein (e.g. silk)
- polyamide (e.g. nylon fibers)
- polyester
- acrylic
- polyolefin (e.g. PP).

The traditional textiles contain fibers that have linear polymer chains, which are to some extent oriented.

Orientation is crucial in fiber production. Orientation means that the macromolecules are aligned in a particular direction or plane and, consequently, enhanced properties are achieved, e.g. enhanced mechanical properties in the alignment direction. In general, there are three principal orientations:²⁴

- uniaxial
- uniaxial by retaining the initial width
- biaxial.

The structure of fibers is characterized by chain orientation with a uniaxial (along the fiber axis) symmetry.⁴³ Fundamentally, fiber formation is simple and is based on steady state uniaxial stretching. However, multiple parameters (process parameters and inherent polymer properties) influence the final fiber structure. Thus, the control over fiber morphology and structure is rather complicated. Parameters, such as polydispersity and molecular weight, have a tremendous impact on the produced fiber properties. High dispersity can result in lower mechanical performance. The optimum process parameters strongly depend on the polymer thermal behavior and crystal nature.

Polymers for fiber manufacturing can be divided in two categories (Figure 1):

- polymers with glass transition temperature (T_g) higher than room temperature (T_R)
- polymers with their T_g lower than room temperature.

The basic difference is that polymers with $T_g > T_R$, which usually exhibit low crystallization rates, are drawn in an amorphous state and need to be heated above T_g to crystallize, while polymers with $T_g < T_R$ can be drawn either at room temperature, or at temperatures higher than their crystallization temperature. Usually, polymers with $T_g < T_R$ are characterized by high crystallization rates. However, polymers of both categories, after, or simultaneously with, drawing are annealed at temperatures between the drawing and the melting temperature for addressing various issues e.g. further increase in crystallinity (besides the crystallinity formed during drawing) or structure stabilization.⁴³

It is worth mentioning, that there is a great demand for fibers from polymers with both T_g lower than room temperature and low crystallization rates. Such category includes biodegradable aliphatic polyesters (with the exception of poly (lactic acid)) and PP.⁴³ As already mentioned, the polymer T_g , plays an important role in selecting the processing (spinning and drawing) conditions. For PP, a rapid cooling after extrusion is preferred in order to obtain a polymer matrix in the paracrystalline – smectic form,⁴³ otherwise α -crystals will be formed prior to drawing and this phenomenon is usually undesirable, since it may result in inhomogeneous stretching. Other factors, which can lead to inhomogeneous deformation during PP drawing, are low drawing temperatures, high drawing speed and higher orientation. During fiber drawing, a temperature increase in the necking region occurs.⁴⁴ For PP fiber production a weight average molar mass (M_w) between

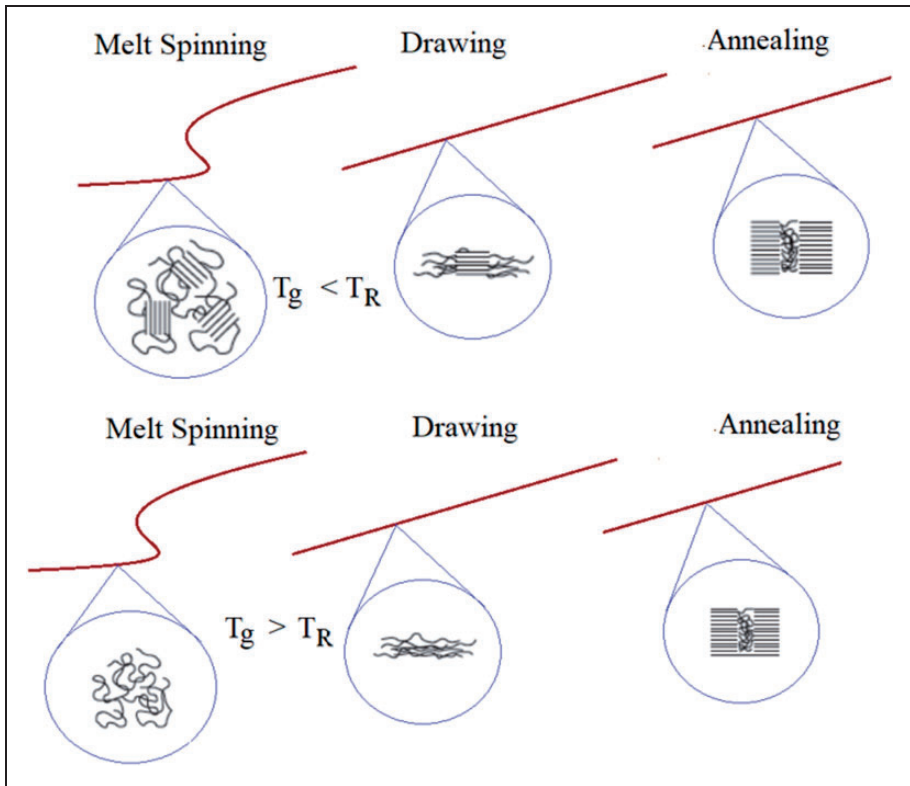


Figure 1. Two polymer fiber manufacturing categories: polymers with glass transition temperature $T_g > T_R$ and polymers with their $T_g < T_R$.

200,000 and 350,000 g/mol is suitable, while PP with M_w up to 600,000 g/mol can be used for monofilament manufacturing.⁴⁵

The melt spinning/drawing process is a well-established technique that is being used for PP fibers production for both research and commercial purposes.^{46–51} Many literature studies aim to improve the melt spinning/drawing process by adding more steps or by altering important process parameters. In this direction, Mukhopadhyay, Deopura and Alagirusamy used a gradient temperature drawing and high modulus high tenacity PP filaments have been prepared. Such filaments were characterized by low shrinkage values even at high temperatures (e.g. 150°C).⁵² Furthermore, a combining melt spinning with wet spinning yields high orientation, but not crystalline morphology in the as-spun fibers.⁵³ Such fibers after a rather mild drawing (ratio up to 1.5) result in PP-fibers with higher, by 9°C, melting point, higher tenacity and modulus. Similar results (oriented amorphous structures with low crystallinity) are reported for high drawing ratios achieved by a gradient drawing system.⁵⁴ In another study, the oriented smectic

phase in fiber and mold grade PP blends, was obtained by melt spinning and drawing at 50 °C. Further drawing these fibers at 145 °C resulted in increased elastic modulus.⁵⁵ i-PP of high tacticity (98%) and ultra-high molecular weight was gel spun and drawn at draw ratios between 4–50. For the gel spinning, a high temperature PP solution in paraffin oil was used and a rapid quenching in water-ice bath was used to obtain a gel fiber.⁵⁶ After drawing, tensile strength increased up to 1.3 GPa. Laser-heated drawing of i-PP fibers has also been reported.^{57,58} Depending on laser power, high drawing ratios (up to 19.5) were achieved and tensile strength up to 910 MPa was measured.⁵⁷ In addition, it is reported that besides laser intensity, the state of polarization of the used laser influences the PP fiber optical and mechanical properties.⁵⁹ Moreover, laser drawing at supersonic velocities yield i-PP nanofibers.⁶⁰

For special applications, bicomponent melt spinning can be used. In this direction, liquid core fibers, which can be used to produce either hollow fibers, or in other applications, e.g. self-healing, were obtained.⁶¹ PP liquid-core (with saturated complex ester-Synative® ES 3345) and hollow fibers have been developed. The hollow fibers exhibit higher thermal stability, while the liquid core fibers degrade at a lower temperature than neat PP.⁶¹ Furthermore, the bicomponent melt spinning can be used to make a core/sheath structure, e.g. PP-based core and polyamide sheath.⁶²

The optical properties and structural deformation of drawn PP fibers were extensively reported in literature.^{63–66} Higher draw ratios yield fibers with higher birefringence as it was revealed by interferometric techniques.⁶⁷ Small angle light scattering is usually applied to study transcrystallization and fiber orientation in PP-based fibers.⁶⁸ A multiple-beam interferometric method was used to study i-PP fiber drawing and it was reported that the drawing velocity has less effect on the molecular structure compared to the draw ratio.⁶⁹ Furthermore, microthermal analysis combined with wide-angle X-ray diffraction were used to evaluate the drawn PP fiber microstructure and melting behavior. It was shown that the PP chains were oriented along the drawn axis, and also 40° to the drawn axis direction.⁷⁰

Finally, in many studies how the specific PP type (i.e. origin of PP, melt flow index, molecular weight) affected the final fiber properties was investigated. In this direction PP-fibers from recycled PP have been produced. Such fibers present increased birefringence and lower mechanical performance.⁷¹ Furthermore, Mao, Wyatt, Chien, Chen, and Yao reported that low molecular weight (high melt flow index) PP is more suitable for the melt spinning process, but the drawing/jet stretching in the melt is difficult.⁷² Consequently, for low molecular weight PP, solid state drawing is a better choice.⁷² In this direction, a solid state two stage hot drawing procedure was used and fibers with 600 MPa tensile strength and 1.2 GPa Young's modulus were obtained.⁷² Polypropylene characteristics also depend on the polymerization process. It was observed that i-PP produced by metallocene catalysts and Ziegler-Natta catalysts yield fibers with different characteristics, i.e. the increase of melting point after drawing is higher

for fibers based on metallocene catalysts.⁷³ Polyhedral oligomeric silsesquioxane based additives have been used in PP composites, which acted as lubricating agents that facilitate drawing.⁷⁴

Electrospinning and other methods. Besides manufacturing conventional microfibers via melt spinning and drawing, nanosized fiber production is possible by electrospinning. Electrospinning is a rather old technique and recently has gained increased interest in the production of nanofibers for biomedical⁷⁵ and other applications.^{76–78} Briefly, in the electrospinning process, a polymer melt or a polymer solution flows through a syringe. An electrical voltage is applied between the needle and an electrically conductive grounded collector. The attractive electric force initially deforms the polymer droplet exiting the needle into a cone shape (Taylor's cone) and, subsequently, stretches the polymer, which forms nanofibers, accompanied by solvent evaporation (solution electrospinning), or polymer solidification (melt electrospinning). In general, melt electrospinning is used for polymers with poor solubility in volatile solvents and is considered as a more environmentally friendly approach. However, higher diameter fibers are usually obtained compared to solution electrospinning.⁷⁹

Electrospinning is widely used to produce PP-based nanofibers. Due to PP's poor solubility, the melt electrospinning approach is quite common.^{79–88} In some cases, and in order to enable the melt electrospinning process, various "lubricants" for lowering the PP melt viscosity,⁸⁹ or conductive additives to increase electrical conductivity, are used.^{89,90}

Other melt electrospinning processes were reported, including needleless melt electrospinning,^{91–94} laser assisted electrospinning,^{95,96} melt electrowriting,⁹⁷ melt differential centrifugal electrospinning⁹⁸ and melt blowing electrospinning.⁹⁹

High temperature and ambient temperature solution electrospinning were used for both i-PP^{100–104} and s-PP.^{82,105–115} i-PP fibers with hierarchically porous structure were produced by combining solution electrospinning and thermally induced phase separation.¹¹⁶

Besides electrospinning other techniques were used to produce PP- nanofibers. PP- nanofibers via melt blowing was reported.¹¹⁷ Also a new technique (ForcespinningTM) based on centrifugal spinning and centrifugal force instead of electrostatic force as in electrospinning, has been successfully used to develop PP nanofibers.^{118–121} Melt spinning PP fibers by cold air attenuation was reported.^{122,123} Finally, porous PP fibers were produced via melt spinning an immiscible PP and polyvinyl alcohol (PVA) blend and subsequently removing the water soluble phase.¹²⁴

Polypropylene additives, fillers and composites

There are very many additives used in PP-based materials. Additive use may be related either to the final application desired properties, e.g. increased electrical resistance,¹²⁵ hydrophobicity,^{82,126} or to PP processing, or both. Additives reduce

the PP content and, thus, the final material cost may be reduced, if the additive is cheaper than PP.¹¹ Usually, each additive serves a specific purpose, e.g. mechanical reinforcement, but inevitably other properties may be altered, e.g. thermal behavior and crystallinity. Besides mechanical and thermal stability reinforcement, additives can alter wettability (in some cases may induce superhydrophobic properties in PP membranes) and other properties, such as roughness, fire retardance, permeability, electrical conductivity, static electricity, optical properties, density (e.g. foams via blowing agents), chemical and biological resistance etc. Furthermore, there are many additives that offer better processability and are not specifically connected to the final product desired properties. In this direction, metal deactivators and antioxidants are used to avoid/constrain thermal oxidation during high temperature processing, lubricants for improving melt flow behavior and slip or antiblock agents for reducing friction of the solid material (fiber, film etc.) through the processing equipment.¹¹

One must distinguish between composites in general and “actual” nanocomposites. For example, multilayered materials are composite materials for specific applications, but usually such composites are not characterized by improved mechanical and thermal properties. On the contrary, typical composites, with dispersed micro- and nano-phases into the polymer matrix, usually exhibit enhanced mechanical strength and thermal stability. In most cases, better dispersion results in more enhanced properties. Poor mixing, however, may result in an inferior composite material compared to the neat polymer. On the other hand, “actual” nanocomposite structures, i.e. nano-scale nanoparticle dispersion with strong interactions and adhesion in the polymer matrix, may result in “dramatic” polymer property changes, even at very low additive loadings (e.g. 0.2 wt%). A representative example is the covalent bond between PP and an additive.¹²⁷

The dispersion of a filler in a polymer matrix, and the subsequent formation of actual nanocomposites, requires sufficiently favorable enthalpic contributions to overcome any entropic penalties.¹²⁷ However, the occurrence, in PP molecule, only of methyl or methylene groups, which interact only by weak dispersive forces with neighboring molecules, results in rather weak intermolecular interactions with fillers. PP cannot take part in hydrogen bonding or strongly polar interactions. Thus, PP, due to its non-polar nature, exhibits poor compatibility with various additives, such as montmorillonite, talc, CaCO₃, carbon nanotubes etc. To overcome such problem, the functionalization of PP and/or the modification of the additive is required.

A typical example of additive modification is the modification of montmorillonite (MMT), which is a phyllosilicate mineral that has been widely used in polymer composites. The alteration of the hydrophilic nature of MMT is accomplished by organic modification (organically modified MMT, OMMT) with alky-surfactants, e.g. alkyl ammonium salts, by ion exchange. Although this approach gives good results for various polymers with polar groups (e.g. polylactic acid), it fails in the case of PP. Since PP and the alkyl chain of the surfactant (modifier) are characterized by the same aliphatic and non-polar nature, they interact by weak dispersion forces, which are not enough to promote extensive dispersion.¹²⁷ Thus, systems are

needed, in which PP-additive interactions are more favorable than surfactant-additive interactions. In this direction, two concepts have been suggested:

- improving the PP-additive interactions
- weakening surfactant-additive interactions.

Functionalizing of PP by inserting polar groups, e.g. random copolymers with p-methylstyrene with and without maleic anhydride, is a common approach of the first route.¹²⁷ Moreover, replacing hydrogen with fluorine in the alkyl chain of the surfactant, in case of organically modified MMT, since the semi-fluorinated chain presents even weaker interactions with MMT than PP, is a characteristic approach of the second route.¹²⁷

PP modifications resulted in intercalated and exfoliated OMMT structures in nanocomposite materials.¹²⁷ Probably, PP grafted with maleic anhydride is the most common, but not in all cases effective, PP and inorganic fillers compatibilizer.^{128–132}

Various fillers with or without modification have been suggested to develop PP micro- and nano- composites. MMT based nanocomposites with enhanced mechanical strength were already mentioned.¹²⁷ Carbonaceous fillers such as graphene have been used. In this direction, exfoliating graphene by thermal expansion was accomplished prior to mixing with PP. Increased thermal stability and electric conductivity, compared to neat PP, were obtained, while for 10 wt% graphene content the modulus increased by 113%.¹³³ Chemically functionalized single wall carbon nanotubes with fluorine led to improved dispersion and adhesion of the fluorinated nanotubes in the PP matrix and to enhanced nanocomposite mechanical properties.¹³⁴ Diethoxy(distearoyl) silane has been used to modify zeolite microparticles. The PP-modified composites exhibited enhanced mechanical strength.¹³⁵

Finally, the all-PP multifilament composites are worth mentioning. The all-PP composites concept uses PP both as matrix and reinforcement agent. The main advantage is that recycling is enabled while PP-composites with additives that are different than PP must be separated prior to recycling.¹³⁶ In this direction, highly oriented drawn PP-tapes have been used to develop fully recyclable polypropylene composite laminates.^{136,137}

Polypropylene based nanocomposite fibers

Depending on the filler, polypropylene based nanocomposite fibers can be arranged in three major categories:^{138–157}

- mineral clay composite fibers
- inorganic nanoparticle composite fibers
- composites with carbonaceous fillers.

Literature studies on such categories, as well as relevant information on the production methods and major conclusions are briefly presented in Table A1 of the Supplemental material file.

Mineral clay composite fibers

As already mentioned, mineral clays are among the most popular fillers to prepare PP nanocomposites and more specifically in PP – nanocomposite fibers, while organically modified montmorillonite clay (OMMT) is probably the most common phyllosilicate additive in polymer composite technology. In this direction, organically modified MMT (Cloisite[®] 15 A) has been used to develop i-PP-MMT nanocomposite fibers using maleic anhydride grafted PP as compatibilizer. A significant mechanical and thermal behavior improvement was achieved at low clay content (0.25–0.5 wt%).¹²⁹ Organically modified montmorillonite (Cloisite[®] 15 A and Dellite 67 G) and a synthetic hydrotalcite modified with hydrogenated fatty acid (Perkalite F100) were used as nanofillers for i-PP using the common compatibilizer (PP grafted with maleic anhydride). Such fibers were characterized by increased thermal stability and similar tensile break compared to neat-PP, despite the fact that higher draw ratios were achieved for the composite fibers.¹²⁸ Another MMT organic modification (Cloisite[®] 20 A) has been used to prepare PP nanocomposite tapes by melt compounding and hot pressing, without any compatibilizer. In this case, the mechanical properties improvement was attributed not to the PP reinforcement, but to the MMT acting as a processing additive that resulted in increased drawing ability.⁴² Cloisite[®] 20 A filler was also used to prepare PP nanocomposite fibers. SEM images showed that nanocomposite fibers presented more uniform surface, more narrow fiber diameter distribution and less roughness.¹³⁸ i-PP/OMMT (C.44 Pa) nanocomposite fibers were produced in order to create co-woven glass-fiber-i-PP/OMMT yarns. Results indicated that low filler concentration resulted in an exfoliated structure, although the process was not optimized.¹³⁹ Montmorillonite clay (MMT), organically modified with cetyl trimethyl ammonium bromide, was used to produce PP/MMT nanocomposite fibers, which showed improved mechanical properties. Young's modulus increased twofold, while tensile strength increased from 532 to 690 MPa.²⁵ Other minerals, such as boehmite, hectorite and sepiolite, have been used as nanosized fillers. For boehmite, an 1 wt% loading resulted in a minor modulus increase.¹⁴⁰ Also, s-PP hectorite composite fibers have been reported. i-PP grafted with maleic anhydride was used to compatibilize s-PP and fluorohectorite modified with octadecylamine.¹⁴¹ This nanofiller, although it increased the melting point of the nanocomposite fibers after drawing (compared to neat s-PP), did not affect the crystallinity. Finally, sepiolite modified with cetyl trimethyl ammonium bromide was used to produce PP multifilament yarns. The composites were characterized by increased thermal stability and improved fire retardant properties.¹⁴²

PP – inorganic nanoparticle composite fibers

Inorganic nanoparticles are another broad filler category. In this direction, nanocomposite PP filaments with silver and silver/TiO₂ nanoparticles have been prepared to develop filaments with permanent antimicrobial efficiency. Depending on

the mixing method, such fillers improved the drawn fiber tenacity and modulus, however there was not observed any considerable crystallinity alteration.^{158–160} In another study, TiO₂ nanoparticles improved mechanical properties and slightly altered fiber crystallinity.¹⁴⁶ i-PP - fumed nanosilica nanocomposites produced by melt spinning and drawn at a 10:1 ratio were characterized by increased elastic modulus.¹⁵¹ Improved i-PP mechanical properties were also obtained with hydrophobic nanosilica filler and maleic anhydride grafted PP compatibilizer.¹³⁰ Fumed nanosilica (12 nm average particle size) grafted with poly(butyl acrylate) and melt mixed with i-PP was used to prepare compression molded uniaxially drawn nanocomposites. A considerable tensile strength improvement (up to 320 MPa) was observed and was attributed to good nanofiller dispersion, which causes stress sharing between nanofiller particles and aligned chains in amorphous regions.¹⁶¹ PP - ZnO nanocomposite fibers (prepared by melt spinning) and non-woven fabrics were produced and showed an increased UV protection and antibacterial activity.¹⁵⁶ BaSO₄ nanoparticles at 40–60 wt% were used to reinforce PP fibers. Dodecyl benzene sulfonic acid was the compatibilizer and the composite fibers exhibited increased thermal stability.¹⁴⁷ PP/CaCO₃ composite nonwoven meshes have been prepared by graft polymerizing poly(acrylic) acid onto PP-fibers surface and subsequent CaCO₃ mineralization on fiber surface. Such PP composite meshes showed superhydrophilic behavior.¹⁰⁰ Melt spun PP-cobalt phthalocyanine complex composite fibers have been produced for potential catalytic removal of mercaptans.¹⁴⁸ PP grafted with poly(ethylene glycol) was used to prepare nanocomposites with gold nanoparticles. The composite electrospun fibers (0.3–2.5 μm) showed rapid catalytic reduction of methylene blue in the presence of sodium borohydride.¹⁶² Finally, using electrospinning, the PP-glass fiber nanocomposite nanofibers have been reported. Their hydrophobicity increased with the amount and PP fiber size.¹⁰⁹

PP composites with carbonaceous fillers

Carbonaceous fillers have been commonly used to prepare electrically conductive polymers.¹⁶³ Carbon nanofiber-PP nanocomposite filaments with high draw ratios (8–19), obtained by gradient drawing, exhibited the same tensile modulus (16.8 GPa) as the neat polymer (16.4 GPa), while the nanocomposite tensile strength was 770 MPa, which was an increase from 670 MPa in the neat PP filament.⁵⁴

Vapor grown carbon fibers and i-PP nanocomposite fibers prepared by melt blending and drawing in 81:1 draw down ratio exhibited a strong stiffening effect due to the aligned carbon fibers and PP chains.¹⁴⁵ Conductive PP textile fibers with graphite and carbon black as fillers have been reported. Addition of the two fillers in equal amounts improved the composite processing properties.⁶²

Single-wall carbon nanotubes (SWNT) - PP fibrous composites were produced by melt-spinning. The study showed that with increasing filler's content, Young's modulus and yield strength increased linearly, while, ductility decreased linearly.¹⁵⁴

McIntosh et al.¹³⁴ compared two different single-wall carbon nanotubes as reinforcement agents in nanocomposite polypropylene fibers. Specifically, they used conventional SWNT and fluorinated SWNT (F-SWNT) and compared their mechanical properties. The authors suggested a reaction mechanism, which results in a covalent bond between the F-SWNT and the polymer matrix. As expected, a covalent bond results in a drastic filler/polymer adhesion improvement, considering that PP is non-polar, resulting in enhanced mechanical properties (tensile strength increased 151.2% and tensile modulus increased 110.5%, for the 10 wt % F-SWNT/PP fibers compared to neat PP fibers).¹³⁴

Multi-wall carbon nanotubes (MWNTs) and PP nanocomposite uniaxial oriented tapes have been produced by melt-based processes and solid state drawing.^{164,165} Low filler content and draw ratios yielded the best results for PP reinforcement. Such tapes exhibited tensile strength up to 500 MPa and high electric conductivity up to 275 S/m.¹⁶⁵ Another study reported that in a melt mixing-melt drawing process, MWNTs at 0.25% increased the PP modulus by 138%.¹⁴³ i-PP microfibers with 0.2% MWNTs, at 5:1 draw ratio, yielded breaking tenacity as high as 1088 MPa.¹⁵² Photothermal (with near infrared laser) PP-MWNTs nanocomposite fiber drawing showed the same PP crystal orientation and a higher MWNT orientation, compared to the furnace-drawn fibers. The photothermally drawn fibers exhibited higher breaking stress and modulus.¹⁵⁵ Higher MWNTs loading (15 wt%) may increase the conductivity, but the melt spinning and drawing processes are hindered, since fiber breakage occurred due to poor mixing.¹⁴⁴ For enhanced mixing, PP grafted with maleic anhydride was used as compatibilizer, while amine groups were added to the MWNT surface. However higher draw ratios (>16) were needed to achieve higher tensile strength, compared to the neat PP fibers at an 8:1 draw ratio.¹³² Oxidized carbon nanotubes were also used in PP composites. The mechanical properties and flame properties deteriorated, and the thermal stability was enhanced.¹⁴² In other studies, MWNTs at low loadings (e.g. 1 wt%) increased crystallinity and only slightly improved tensile strength.^{149,157}

The addition of nonionic surfactant improved the MWNT dispersion in a PP-decalin solution that was used to produce nanofibers through the ForcespinningTM technique.¹¹⁹ i-PP/MWNTs nanocomposite fibers, produced by high temperature solution electrospinning, exhibited high thermal stability and crystalline memory effect.¹⁰⁴ PP-MWNTs microfibers (1–3 μm) have been prepared by melt electrospinning with a small paraffin liquid addition for dispersion and spinnability improvement. Such membranes were characterized by increased conductivity and hydrophobicity (water static contact angle 120–130°).¹⁵⁰

Another carbon-based additive that has been used for PP nanocomposite fiber development is graphene nanoplatelets (GnPs). PP nanocomposite fibers with 0.2 wt% GnPs showed increased thermal stability compared to neat PP fiber.¹⁵³

Oriented wood-PP composites were also developed with 40 wt% wood loading. In non-drawn samples stiffness increased from 1.9 to 8.2 GPa and

strength increased from 13 to 127 MPa. However, the drawn composites exhibited lower modulus. In addition, the use of different wood types and the use of maleic anhydride as compatibilizer did not significantly influence their mechanical properties.¹³¹

Applications of PP fibers

Polyolefin fiber is defined as any fiber in which the fiber-forming substance is more than 85% in ethylene, propylene or other olefin units.⁴⁵ The most popular polyolefin fibers, by far, are based on PP, while, as already mentioned in the introduction section, the PP fiber-based products require about 1/3 of global PP production.

PP fibers present characteristic properties, which define to a large extend their applications. In more detail, drawn fibers exhibit increased crystallinity and are characterized by increased strength and elastic modulus, which are proportional to the draw ratio. In essence, due to the chain alignment, the same force can be tolerated in a smaller cross section resulting in increased stress. By increasing orientation, the elongation at break decreases. Orientation also enhances their low temperature mechanical strength. Oriented films and fibers are tough at temperatures below zero, while without orientation they are brittle.¹²⁵ For continuous use, the maximum operating temperature for PP is around 100 °C.¹¹

Orientation is applied to all PP fiber applications and 75% of the PP films are biaxial oriented films.¹²⁵ However, fibers and in general stretched structures, due to their oriented chains, present a “memory behavior”, i.e. if reheated at temperatures high enough and close to the orientation temperature, they tend to return to their initial shape. Such memory effect can be considered as an advantage for developing heat-shrinkable films. On the other hand, it is unwanted in many applications, since it relates to poor high temperature stability, which limits their potential applications.¹²⁵ Shrinkage also occurs, but slowly, even at room temperatures. To avoid such problem, annealing or a controlled shrinkage (~ 5%) at high temperatures (even at higher than the orientation temperature) is applied. Such controlled shrinkage is usually applied in line with the orientation process. High orientation temperatures can also reduce shrinkage.

Finally, PP fibers are skin friendly, non-allergenic and in general hydrophobic,¹²⁶ while depending on the fabrication technique, they may exhibit superhydrophobic behavior.⁸² Furthermore, biaxial oriented PP films, besides enhanced mechanical properties, exhibit very low water vapor permeability and increased electrical resistance.¹²⁵

Due to such characteristic properties, PP presents many applications including household applications (bowls etc), automobile and other industries (e.g. washing machine parts), packaging, piping, furniture and, of course, fiber based applications.¹¹ Conventional PP-based fiber applications include artificial surfaces (e.g.

for sport activities), rope and cordage, carpet backing, packaging sacks and tarpaulins.¹¹ In such applications drawn PP and PP nanocomposite fibers are used. PP-based fibers are also used in surgery masks¹²⁶ and have been suggested for air filtration.¹⁶⁶ Finally, drawn fibers from virgin and recycled PP have been used for concrete reinforcement replacing steel mesh.¹⁶⁷

Nevertheless, the recent research interest in electrospinning technique, resulted in novel PP fiber applications. Since nanofibrous electrospun membranes do not exhibit strong mechanical strength, they are often formed on microporous meshes. PP meshes are commonly used as substrates. The electrospun membrane adhesion on PP mesh can be enhanced by plasma treatment.¹⁶⁸ Electrospun PP fibrous oriented membranes have been suggested for water filtration with increased efficiency, with similar permeate flux as the one of random fiber membrane.⁹³ Superhydrophobic PP micro- and nano-fibrous membranes have been prepared by electrospinning. Coating such electrospun PP fibers on nonwoven PP mesh improved barrier performance with potential application as protective clothing or battery separators.⁸² Microporous PP meshes are used in battery separators as barriers for the electrode contacts, and also act as electrolyte's reservoir. A recent trend is coating the PP meshes with nanofibers from other polymers, e.g. (polyvinylidene fluoride), to improve the PP mesh performance.¹⁶⁹ PP electrospun membrane coated with a PVA electrospun membrane has also been suggested for water filtration.⁹⁴ Electrospun fibrous PP membranes have been used as base for sensor development to detect trihalomethanes in water.¹¹⁵

Other fibrous PP membrane applications that present superhydrophobic properties include separating water droplets from diesel fuel.¹¹² In addition, melt electrospun PP fibers have been found effective for cleaning up marine oil spills.^{80,92} Electrospun or melt blown microfibrillar PP hydrophilic (by in situ CaCO_3 formation) nanocomposite meshes have been suggested for water/oil separation.¹⁰⁰ In general, in some applications the PP hydrophobicity is a limiting factor, thus hydrophilization can broaden the applications (e.g. disposable baby diapers, surgical dressings etc.).¹¹⁸

Composite PP fibers except for mechanical and thermal behavior improvement, have been used as catalyst-supports (e.g. for removing mercaptans from aqueous solutions).^{148,162} Electrically conductive PP fibers, mainly produced by adding MWNTs,^{150,165} widen the potential PP fiber applications, e.g. in electrochemical and electronic devices.

Grafting PP electrospun nanofibers with suitable agents also expands the PP fiber applications. Amine functionalized PP nanofibers have been suggested as high potential CO_2 absorbents.¹⁰⁶ Also, amino-acid grafted PP nanofibers have been proposed for antimicrobial peptide binding medical clothing.¹⁷⁰

Conclusions

The traditional melt/spinning drawing process yields fibers in the microscale, while other techniques, such as electrospinning, allow the production of nanofibers.

Solution electrospinning is more easily applied to s-PP, since it is easier to dissolve than i-PP.

However, the microfibers produced by the traditional melt spinning/drawing process are characterized by a considerable increase in tensile strength, e.g. from 30 MPa in the non-drawn sample to more than 300 MPa in the drawn fibers. Such considerable mechanical performance enhancement can be further enhanced by adding fillers for preparing composite drawn fibers. However, PP presents only weak intermolecular interactions with most inorganic fillers due to its non-polar nature, which hinders successful dispersion of most nanoparticles in the polymer matrix. Consequently, simply mixing PP and fillers is usually not enough and modifying PP, or the filler, or both, is required. Grafting polar groups onto PP chains improves interaction with polar fillers. Very commonly, PP grafted with maleic anhydride is used as compatibilizer in PP-composites. However, the composite fiber properties often present only marginal improvement. Notably, covalent bonding between PP chains and functionalized carbon nanotubes significantly contributes to improved mechanical performance.

Besides tensile strength, other PP properties, such as electrical conductivity and hydrophobicity, can be tuned by adding appropriate fillers. Such property alteration expands PP's applications. Some new applications, such as CO₂ absorption or water/oil separation, seem to be attractive, if PP's low cost and recyclability are taken into consideration. However, developing composites based on recycled PP is rather limited and, consequently, future research may focus on suitable compatibilizers/additives for such recycled polymer matrices.

Finally, polymeric materials that have oriented chains can find applications in shape memory materials. Although, oriented PP can be a heat shrinkable film, utilizing drawn fibers as a shape memory material is rather limited. In composite materials, using a filler that depends on the non-covalent interactions between polymer and filler, as in the case of polyurethane,^{171,172} provides memory shape effect induced by other factors than heat, e.g. light or electric current. Thus, the PP drawn fiber shape memory disadvantage might be tuned by appropriate fillers and lead to innovative applications.

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